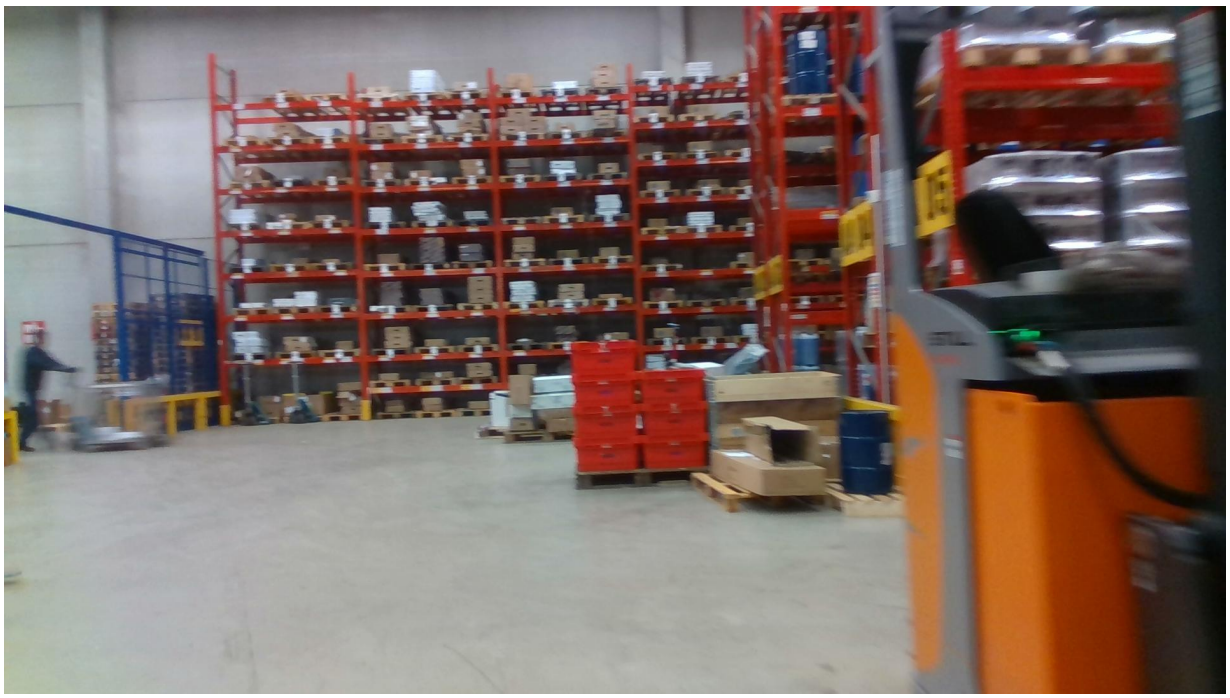
# Primerjava modelov VLM in LLM na industrijskih slikah

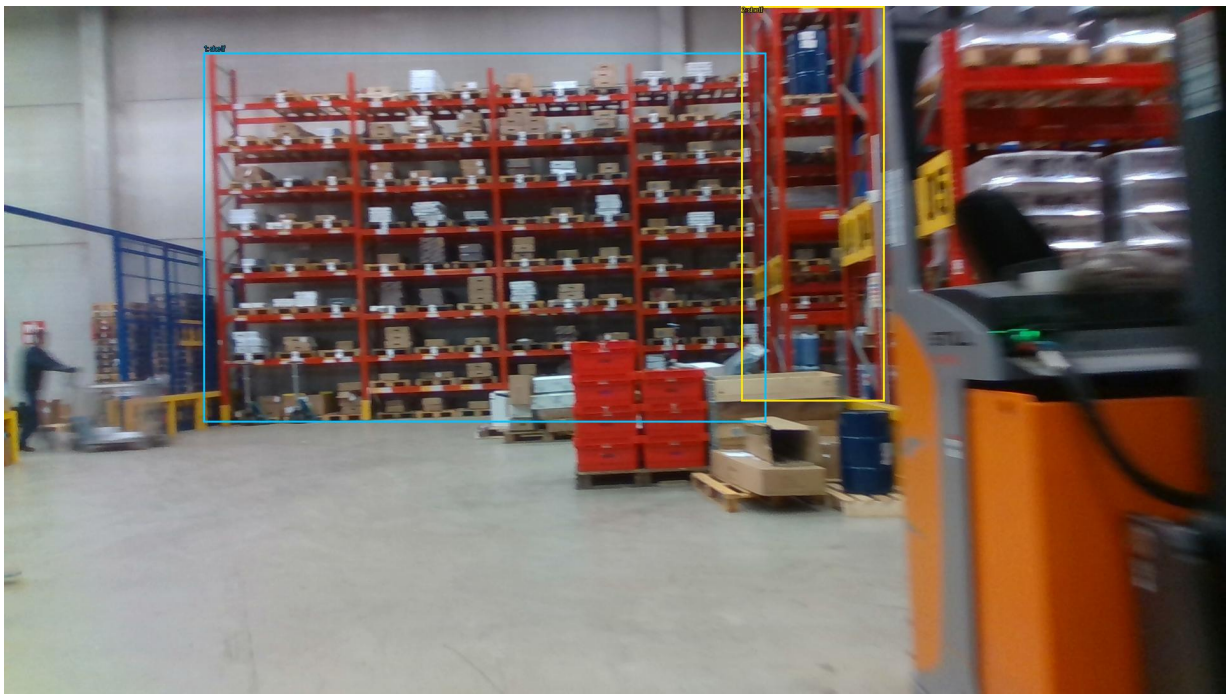
## 1 VLM

VLM prejme fotografijo skladišča in vrne opis scene. Florence-2-large uporablja tudi namensko zaznavanje objektov (OD). Qwen3-VL-8B in NVILA-8B sta bila dodatno preizkušena z zahtevo za koordinate okvirjev. Besedilni opisi so iz prvega testa, okvirji pa iz dodatnega testa. Veljaven okvir še ne pomeni pravilno zaznanega objekta.

### 1.1 loco-s3-36540



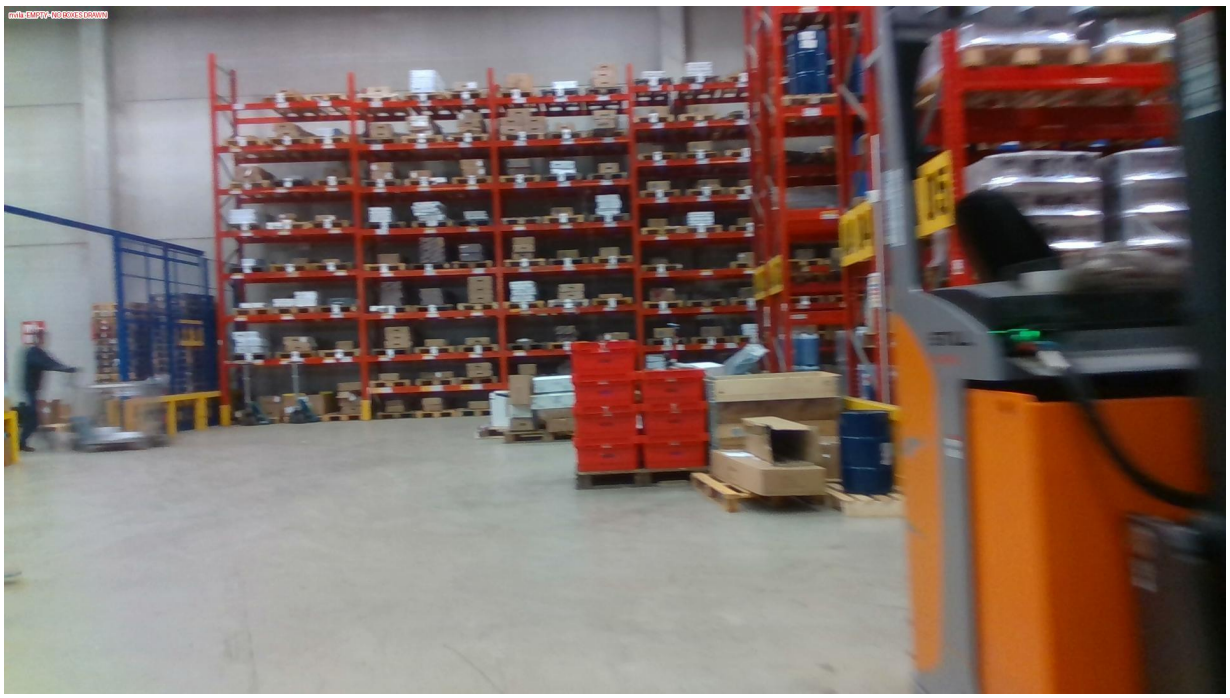
Slika 1: Originalna slika



Slika 2: Florence – okvirji



Slika 3: Qwen – okvirji



Slika 4: NVILA – okvirji

**Florence – opis:** The image shows a large warehouse with rows of red metal shelves. The shelves are filled with various items such as boxes, pallets, and other storage containers. The floor is made of concrete and there is a forklift on the right side of the image. The warehouse appears to be well-maintained and well-organized. The image is taken from a low angle, looking up at the shelves.

**Qwen – opis:** A wide-angle view of a large warehouse interior with high ceilings and concrete floors. Dominant red metal shelving units fill the background, stocked with boxes and pallets. An orange forklift is partially visible on the right. A person walks in the left distance near blue racks. The scene suggests active inventory management.

**NVILA – opis:** The image depicts the interior of a warehouse or storage facility. The space is filled with tall, red metal shelving units arranged in rows, each unit containing multiple levels of storage. These shelves are densely packed with various items, including boxes and containers of different sizes and colors. The floor is concrete, and there are several pallets scattered throughout the area, some stacked on top of others. On the right side of the image, there's an orange machine with a black arm, possibly a forklift or similar equipment used for moving heavy objects. In the background, there's a blue metal gate, and a person wearing a dark jacket can be seen walking towards it. The overall atmosphere suggests a busy, active workspace where goods are stored and organized for distribution or inventory management.

*Komentar:* V dodatnem testu je Qwen vrnil sedem veljavnih okvirjev. Florence je lokaliziral predvsem regale, NVILA pa ni vrnila okvirjev.

## 1.2 loco-s4-31835




Slika 5: Originalna slika




Slika 6: Florence – okvirji




Slika 7: Qwen – okvirji



Slika 8: NVILA – okvirji

**Florence – opis:** The image shows a large warehouse with rows of pallets and shelves. The pallets are arranged in a neat and orderly manner, and the shelves are filled with boxes of various sizes and colors. The boxes are stacked on top of each other and have the word ”POTT”written on them in white letters. The warehouse appears to be

well-stocked with various items, and there are other shelves and racks visible in the background. The floor is made of concrete and the ceiling is high with overhead lights. The overall atmosphere of the warehouse is industrial and organized.

**Qwen – opis:** A large warehouse interior with high ceilings and metal shelving units. Rows of pallets stacked with white boxes labeled 'POTT' line the aisle. Red boxes are visible on a nearby pallet. The floor is concrete, and yellow safety barriers separate storage areas. Lighting fixtures hang from the ceiling.

**NVILA – opis:** The image depicts a warehouse storage area with rows of pallets and shelves filled with boxes. The floor is concrete, and the ceiling has exposed beams. The warehouse appears to be well-lit, with fluorescent lights visible in the background. The main focus of the image is on several stacks of white boxes labeled "POTT"that are arranged on wooden pallets. These boxes are neatly stacked, creating a uniform appearance. Other boxes can be seen on the shelves in the background, including some red ones. The overall scene suggests an organized storage facility for goods or products.

*Komentar:* NVILA je vrnila en veljaven okvir za regale. Qwenov odgovor s koordinatami ni bil veljaven. Florence je označil le manjši zaboj.

# 2 LLM

LLM ne vidi slike: prejme samo opis in seznam objektov iz VLM. Za vsak objekt določi `fixed`, `movable` ali `uncertain` in vrne razlog. Uporabljen je Qwen3.5 4B BF16 na vhodu Qwen3-VL (prva ponovitev, seme 42). Spodaj so vsi štirje dejanski odgovori na kratki in dolgi poziv brez primerov.

## 2.1 loco-s3-36540: kratek poziv

Odgovor je **neveljaven**, ker četrti objekt vsebuje `description` namesto `reason`.


```
{"objects":[
{"id":1,"classification":"movable","reason":"Tall freestanding shelves can be relocated without
    dismantling."},
{"id":2,"classification":"movable","reason":"Forklift is a standard movable vehicle in warehouse use
    ."},
{"id":3,"classification":"movable","reason":"Person is inherently mobile and not fixed to the
    structure."},
{"id":4,"classification":"movable","description":"Freestanding racks can be pushed or repositioned
    normally."},
{"id":5,"classification":"movable","reason":"Stacked crates on pallets are easily moved by forklift
    ."},
{"id":6,"classification":"movable","reason":"Wooden pallets are lightweight and portable items."},
{"id":7,"classification":"movable","reason":"Cardboard boxes are light, packable, and easily relocated
    ."},
{"id":8,"classification":"movable","reason":"Barrel on a pallet is easily carried or pushed away."}]}
```


## 2.2 loco-s3-36540: dolg poziv


```
{"objects":[
{"id":1,"classification":"movable","reason":"Shelves stand independently with no visible permanent
    mounting."},
{"id":2,"classification":"movable","reason":"Forklift is a vehicle designed to be moved by operators
    ."},
{"id":3,"classification":"movable","reason":"Person is an animate object that moves freely."},
{"id":4,"classification":"movable","reason":"Racks appear freestanding with no visible structural
    attachment."},
{"id":5,"classification":"movable","reason":"Crates are stacked items easily repositioned or moved."},
{"id":6,"classification":"movable","reason":"Pallets are standard warehouse objects designed for
    transport."},
{"id":7,"classification":"movable","reason":"Boxes are lightweight and can be moved by hand or
    machinery."},
{"id":8,"classification":"movable","reason":"Barrel sits on a pallet, making it easily repositionable
    ."}]}
```


*Komentar:* Dolg poziv je vrnil pravilen format. Oba odgovora pa regale označita za premične, čeprav iz same fotografije ni mogoče zanesljivo potrditi njihove pritrditve.

## 2.3 loco-s4-31835: kratek poziv


```
{"objects":[
{"id":1,"classification":"movable","reason":"Stacked boxes on pallets can be easily relocated."},
{"id":2,"classification":"movable","reason":"Pallets are standard warehouse items designed to be moved
    ."},
{"id":3,"classification":"fixed","reason":"Shelving units have visible floor anchors and structural
    integration."},
{"id":4,"classification":"fixed","reason":"Barriers show visible installation evidence and base
    attachments."},
```

{"id":5,"classification":"movable","reason":"Boxes rest on pallets and are not permanently attached
    ."},
{"id":6,"classification":"uncertain","reason":"Floor is a surface, not an object requiring relocation
    classification."}]}

## 2.4 loco-s4-31835: dolg poziv

{"objects":[
{"id":1,"classification":"movable","reason":"Cardboard boxes are easily stacked or moved in normal
    warehouse operations."},
{"id":2,"classification":"movable","reason":"Pallets are standard movable units used for transporting
    goods."},
{"id":3,"classification":"fixed","reason":"Shelving units have visible floor anchors and require
    dismantling to move."},
{"id":4,"classification":"movable","reason":"Safety barriers can be pushed or repositioned without
    permanent installation."},
{"id":5,"classification":"movable","reason":"Cardboard boxes are easily stacked or moved in normal
    warehouse operations."},
{"id":6,"classification":"fixed","reason":"Concrete floor is a structural building element permanently
     part of the facility."}]}

*Komentar:* Oba odgovora sta veljavna, vendar se razlikujeta pri rumenih zaščitah in betonskih tleh. Sprememba poziva torej vpliva na odločitev modela.

## 2.5 Ostali LLM modeli

Isti vhod Qwen3-VL r1; temperatura 0.3, seme 42, ponovitev r1. Izvirni odgovori so brez popravkov.

### 2.5.1 loco-s3-36540

**Qwen3.5 Q4 – Kratki poziv** Status: error. Object 4 has missing reason.


```
{"objects":[{"id":1,"classification":"movable","reason":"Standalone shelving unit can be moved by
    workers"},{"id":2,"classification":"movable","reason":"Forklift is a vehicle designed for
    repositioning"},{"id":3,"classification":"uncertain","reason":"Person classification does not
    determine object status"},{"id":4,"classification":"movable","description":"Freestanding racks can
     be relocated if needed"},{"id":5,"classification":"movable","reason":"Stacked crates are easily
    moved by workers"},{"id":6,"classification":"movable","reason":"Pallets are standard movable
    storage units"},{"id":7,"classification":"movable","reason":"Boxes are lightweight and easily
    repositioned"},{"id":8,"classification":"movable","reason":"Barrels are portable containers on
    pallets"}]}
```


**Qwen3.5 Q4 – Dolgi poziv** Status: error. Object 4 has missing reason.


```
{"objects":[{"id":1,"classification":"movable","reason":"Independent shelving units can be moved or
    repositioned."},{"id":2,"classification":"movable","reason":"Forklifts are designed to be driven
    and relocated."},{"id":3,"classification":"movable","reason":"People move freely and do not stay
    in one place."},{"id":4,"classification":"movable","description":"Freestanding racks can be moved
    or repositioned."},{"id":5,"classification":"movable","reason":"Stacked crates are easily moved by
     forklifts or humans."},{"id":6,"classification":"movable","reason":"Pallets are standard movable
    items in warehouse use."},{"id":7,"classification":"movable","reason":"Boxes can be stacked,
    carried, and relocated freely."},{"id":8,"classification":"movable","reason":"Barrels on pallets
    can be easily moved by forklifts."}]}
```


**Gemma 3 Q4 – Kratki poziv** Status: ok.


```
{"objects":[{"id": 1, "classification":"fixed", "reason":"Tall metal shelves are permanently installed
     in the warehouse."}, {"id": 2, "classification":"movable", "reason":"Forklift is a vehicle that
    can be moved around."}, {"id": 3, "classification":"movable", "reason":"Person is a human and can
    move freely."}, {"id": 4, "classification":"fixed", "reason":"Blue metal racks are permanently
    installed in the warehouse."}, {"id": 5, "classification":"movable", "reason":"Plastic crates are
    carried and repositioned during inventory."}, {"id": 6, "classification":"movable", "reason":"
    Wooden pallets are used to transport and store items."}, {"id": 7, "classification":"movable", "
    reason":"Cardboard boxes are moved and stacked for storage."}, {"id": 8, "classification":"movable
    ", "reason":"Blue barrel is a container that can be carried."}]}
```


**Gemma 3 Q4 – Dolgi poziv** Status: ok.


```
{"objects": [{"id": 1, "classification": "movable", "reason": "Shelving units are movable and
    repositionable."}, {"id": 2, "classification": "movable", "reason": "Forklifts are designed for
    transport and movement."}, {"id": 3, "classification": "movable", "reason": "People move around
    within the warehouse."}, {"id": 4, "classification": "movable", "reason": "Racks can be moved or
    repositioned easily."}, {"id": 5, "classification": "movable", "reason": "Crates are designed to
    be carried and stacked."}, {"id": 6, "classification": "movable", "reason": "Pallets are used for
    transporting and storing goods."}, {"id": 7, "classification": "movable", "reason": "Boxes can be
    moved and rearranged within the space."}, {"id": 8, "classification": "movable", "reason": "
    Barrels are designed to be moved around."}]}
```


Komentar: Gemma spremeni klasifikacijo regalov; Qwen Q4 vrne nepravilno polje pri objektu 4.

### 2.5.2 loco-s4-31835

**Qwen3.5 Q4 – Kratki poziv** Status: ok.

{"objects":[{"id":1,"classification":"movable","reason":"Boxes rest on pallets and can be easily relocated."},{"id":2,"classification":"movable","reason":"Pallets support boxes and are designed to be moved."},{"id":3,"classification":"fixed","reason":"Shelving units have visible floor anchors requiring dismantling."},{"id":4,"classification":"uncertain","reason":"Barriers show installation evidence but mobility unclear."},{"id":5,"classification":"movable","reason":"Boxes rest on pallets and can be easily relocated."},{"id":6,"classification":"fixed","reason":"Floor is structural base, not relocatable without demolition."}]}

**Qwen3.5 Q4 – Dolgi poziv** Status: ok.

{"objects":[{"id":1,"classification":"movable","reason":"Stacked boxes can be easily moved or removed."},{"id":2,"classification":"movable","reason":"Pallets are standard warehouse items designed to be moved."},{"id":3,"classification":"fixed","reason":"Shelving units anchored to the floor require dismantling to move."},{"id":4,"classification":"movable","reason":"Safety barriers can be relocated by removing their bases."},{"id":5,"classification":"movable","reason":"Cardboard boxes are lightweight and easily repositioned."},{"id":6,"classification":"fixed","reason":"Concrete floors are structural elements that cannot be moved."}]}

**Gemma 3 Q4 – Kratki poziv** Status: ok.

{"objects":[{"id": 1, "classification":"movable", "reason":"Stacked boxes on pallets, can be carried or repositioned"},{"id": 2, "classification":"movable", "reason":"Wooden pallets used to support the boxes, movable"},{"id": 3, "classification":"fixed", "reason":"Metal shelving units are a permanent installation"},{"id": 4, "classification":"fixed", "reason":"Yellow safety barriers are a permanent installation"},{"id": 5, "classification":"movable", "reason":"Red cardboard boxes on a pallet, can be carried"},{"id": 6, "classification":"movable", "reason":"Concrete floor is the ground surface, movable"}]}

**Gemma 3 Q4 – Dolgi poziv** Status: ok.

{"objects": [{"id": 1, "classification": "movable", "reason": "Boxes are stacked on pallets and can be moved."}, {"id": 2, "classification": "movable", "reason": "Pallets are transportable items within the warehouse."}, {"id": 3, "classification": "fixed", "reason": "Shelving units are a permanent structural installation."}, {"id": 4, "classification": "fixed", "reason": "Safety barriers are permanently installed for safety reasons."}, {"id": 5, "classification": "movable", "reason": "Boxes can be moved and are not fixed installations."}, {"id": 6, "classification": "fixed", "reason": "Concrete floor is a permanent ground surface."}]}

Komentar: Razlike v klasifikaciji rumenih zascit in betonskih tal.

# 3 Primerjava rezultatov

V tabelah je locena tehnicna veljavnost izhoda od pravilnosti zaznavanja in klasifikacije. Brez rocno potrjenega ground truth ni mogoce izracunati zanesljive tocnosti.

## 3.1 VLM

| Model | Strukturirani izhod | Opazanje |
|---|---|---|
| Florence-2-large | 90/90 OD odzivov | Najbolj zanesljiv izris koordinat v dosedanjem postopku; pomembne objekte vcasih izpusti. |
| Qwen3-VL | 90/90 opisov | Najbolj uporaben podroben opis scene med pregledanimi primeri; locena bbox naloga je uspela le pri eni od stirih slik. |
| NVILA-8B | 0/90 popolnoma veljavnih JSON opisov | Besedilo je pogosto uporabno, izluscenje pa nezanesljivo. Bbox: en veljaven rezultat od stirih. |

Primerjava koordinat Qwen in NVILA temelji na samo stirih slikah, zato iz nje ne sklepamo o splosni natancnosti lokalizacije. Veljavnost Florence je tehnicna veljavnost OD, ne pravilnost okvirjev.

## 3.2 LLM

Spodaj je delez shematsko veljavnih odgovorov med dejansko izvedenimi klici v dosedanjem agregiranem pregledu (skupaj vec slik, VLM vhodov in ponovitev). Preskoceni vhodi niso vsteti. Veljavnost ni enaka pravilni klasifikaciji.

| Model | Kratek poziv |
|---|---|
| Qwen3.5 4B BF16 | 659/756 (87,2 |
| Qwen3.5 4B Q4 | 681/756 (90,1 |
| Gemma 3 4B Q4 | – |

Za Gemmo zakljucni agregat ni bil ponovno izracunan; stare vmesne stevilke zaradi nepopolnega obsega niso primerljive in jih zato ne navajamo. Po veljavnosti formata med primerljivima Qwen razlicicama vodi Qwen3.5 Q4 pri kratkem in dolgem pozivu. Dolgi poziv je pri obeh Qwen razlicicah zanesljivejsi. Gemma pri pregledanih primerih vcasih pravilneje klasificira regale, a lahko napacno oznaci betonska tla; najboljsi model po dejanski pravilnosti zato se ni dolocen.